

Hypothesis, the T-bill rate reflects *expected* inflation. Movements in expected inflation account for a large part of overall movements in total inflation because of inflation's persistence—if inflation is high today, it is likely to be high next month.

Investors need to be concerned about two things in holding cash as an inflation-hedging tool. First, Figure 11.3 shows that cash returns *lag* inflation. This is most clearly seen in the huge spike in inflation in the late 1970s; the increase in T-bill yields did not come until the early 1980s. It is impossible for a policy authority to respond with perfect foresight to contemporaneous inflation shocks given that all our macro data is collected with a lag.⁴ But once inflation shocks occur, inflation is persistent. Inflation shocks are thus reflected in T-bill yields with a lag. Unless inflation is extraordinarily high (hyper-inflation has never occurred in the United States and is extremely unlikely despite the doomsayers), this lag is inconvenient and modestly costly, but it is not a reason to avoid cash as a real asset.

The second reason is. At the time of writing and at end of the sample in Figure 11.3, T-bill returns were effectively zero. This is exactly where the Fed wanted short-term interest rates to be at this time: the economy was still tender after the global financial crisis of 2008 and 2009 and the Great Recession that followed. Inflation was 3.0% at December 2011 and thus lay square in the middle of the Fed's preferred 2% to 4% policy range. The real interest rate, which is the difference between the T-bill yield and inflation, was negative, and significantly so. Reinhart and Rogoff (2011) and Reinhart and Sbrancia (2011) label this *financial repression*. It is effectively a tax imposed on bond (and cash) investors by policy-makers who channel the benefits to other purposes and people (like shoring up the financial system and subsidizing managers and owners of large financial institutions). Given the ocean of government debt sloshing around worldwide, financial repression is likely to stay a while. This second reason is a statement about the relatively low long-run return of T-bills and that an investor may desire other assets with higher long-run average returns, not that T-bills are not a good inflation hedge.

4. Real Bonds

A real bond is a bond whose principal or coupon is indexed to inflation. In this way the investor is protected against inflation because the payouts grow as inflation increases. Bond payments, therefore, remain constant in real terms. Real bonds are called *linkers*.

The U.S. version of linkers is *Treasury Inflation-Protected Securities* (TIPS), which have been issued by Uncle Sam since 1997. As of December 31, 2011, the

⁴“Forecasting” current economic conditions is called *nowcasting*, and econometricians have developed sophisticated models to do this. See Giannone, Reichlin, and Small (2008).